

Literature Review

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Literature Review 01

Supporting Productivity Skill Development in College Students through Social Robot Coaching: A Proof-of-Concept

Overview:

Lalwani and Salam (2025) developed a Socially Assistive Robot (SAR) as a productivity coach for college students. The main idea of the paper is to help students improve important productivity skills such as time management, task prioritization, self-awareness and motivation.

The proposed system provides six lessons related to time management and task prioritization. Students interact with the system through a chat interface, while the robot can provide responses through speech. A dashboard is also used to monitor the student's progress, mood, engagement, confidence and time spent on different lessons. The system then provides personalized productivity insights to encourage students to reflect on their own study habits. The researchers evaluated the system with 15 college students and reported a System Usability Scale score of 79.2 along with positive ratings for overall experience and engagement.

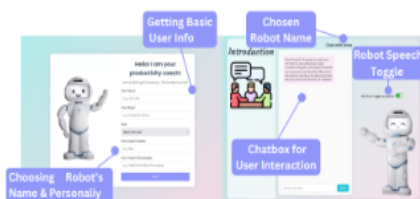


Fig. 1: On Left: Home Page. On Right: Introductory Interaction Page.



Fig. 4: Dashboard Page Layout.



Fig. 5: Experimental Setup.

project Objective:

1. Help students improve their time-management skills.
2. Support students in prioritizing their tasks.
3. Provide interactive productivity coaching instead of only giving static instructions.
4. Track students' progress, mood, engagement and confidence.
5. Provide personalized productivity insights.
6. Encourage students to develop self-awareness and better productivity habits.
7. Investigate whether a social robot can provide productivity coaching in a scalable way.

Research Gap:

Existing productivity applications such as calendars, task-management tools and focus applications can help students organize their activities. However, these systems generally depend on the student's own discipline and do not provide personalized educational coaching.

Productivity books can provide more detailed guidance, but they are usually one-size-fits-all and do not interact with the student according to their individual situation. On the other hand- human productivity coaching can provide personalized support but requires considerable resources and is difficult to provide to a large number of students. Although socially assistive robots have already been used for education, coaching and student support, the authors identify a specific gap: the use of SARs for developing productivity-related soft skills among college students remains underexplored. Another important gap identified by the paper is the use of Large Language Models in this area. The authors state that, to their knowledge LLM had not been formally evaluated for productivity coaching or integrated into a socially assistive robot specifically for this purpose.

Shortly we can say:

There is a need for an interactive and personalized system that can provide productivity coaching to college students while combining the engaging interaction of a social robot with individualized productivity support.

Research Questions:

- 1: Can a socially assistive robot provide useful productivity coaching to college students?
- 2: How effectively can a social robot help college students improve their time-management skills?

- 3: How can a social robot support students in learning effective task-prioritization strategies?
- 4: How do college students perceive the usability and overall experience of a robot-based productivity coaching system?
- 5: Can tracking progress, mood, engagement and confidence help provide more personalized productivity insights?
- 6: How engaged are students when receiving productivity coaching through a social robot?
- 7: How does robotic productivity coaching compare with students' expectations of traditional human coaching?

Key Contribution:

The main contribution of this paper is the development of a proof-of-concept socially assistive robot for college-student productivity coaching. The system combines productivity lessons, interactive communication and a monitoring dashboard to provide personalized feedback.

The researchers tested the system with 15 college students and obtained a SUS score of 79.2 while participants also reported positive overall experience and engagement. However, some participants found typing their responses tiring, and the study reported that robotic coaching was still perceived as somewhat below human coaching by several participants.

Literature Review 02

Ethically-Aware Participatory Design of a Productivity Social Robot for College Students

Overview:

Lalwani and Salam (2025) studied how a Socially Assistive Robot can be designed to support the productivity of college students. The main focus of this paper is not only the technical development of the robot, but also understanding what students actually need from a productivity robot.

The researchers used a Participatory Design approach, where college students and a Student Success and Well-Being Coach were directly involved in the design process. Through interviews and a collaborative workshop, they investigated students' productivity problems, preferred robot features and suitable robot characteristics. The study also considered ethical issues from the beginning so that the robot could support students without unnecessarily reducing their privacy or independence.

Project Objective:

The main objective of the project is to understand the productivity challenges faced by college students and use their feedback to design an effective and responsible productivity-focused social robot.

1. Identify the major productivity challenges experienced by college students.
2. Understand what features students expect from a productivity social robot.
3. Identify suitable physical characteristics and interaction styles for the robot.
4. Understand how students want the robot to support their productivity.
5. Investigate students' expectations regarding personalized robotic assistance.
6. Identify important ethical concerns related to productivity-monitoring robots.
7. Develop design recommendations for future productivity-focused social robots.

Research Gap:

Many productivity tools are already available, including planners, mobile applications and other digital systems. However, these tools generally require students to maintain regular usage and self-discipline. The authors argue that this can be difficult for students who already struggle with productivity and executive-function challenges. Although socially assistive robots have been used in areas such as education, cognitive development and mental health, there is comparatively

limited research on productivity-focused SARs specifically designed around college students needs.

Another important gap is the lack of sufficient attention to student participation and ethical considerations when designing such systems. Since a productivity robot may observe or respond to a student's behavior, issues such as privacy, transparency, user control and empowerment become important

major research gap as:

There is a need for a student-centered and ethically responsible design approach for productivity social robots that directly considers college students' needs, preferences, privacy and independence.

Research Questions:

- 1: What are the major productivity challenges faced by college students?
- 2: What features do college students expect from a productivity-focused social robot?
- 3: What type of physical appearance and interaction style would students prefer in a productivity robot?
- 4: How can a social robot provide productivity support without becoming intrusive or distracting?
- 5: What ethical concerns should be considered when designing a robot that supports or observes students' productivity?
- 6: How can students' participation in the design process improve the usefulness and acceptance of a productivity social robot?
- 7: How can a productivity robot support students while maintaining their independence, privacy and control over their own activities?

Key Contribution:

The main contribution of this paper is a student-centered participatory design approach for productivity-focused social robots. Instead of designing the robot only from the developers'

perspective, the researchers collected feedback directly from students and a Student Success and Well-Being Coach.

The paper also provides ethical guidelines and design recommendations for future productivity robots. This is particularly important because a productivity robot may interact with students' personal routines and potentially observe their activities.

Literature Review 03

Design and Evaluation of a Socially Assistive Robot Schoolwork Companion for College Students with ADHD

Overview:

This paper presents a Socially Assistive Robot designed as a schoolwork companion for college students with ADHD. The main idea is to use a robot's physical presence and subtle movements to provide companionship and support while students complete their academic work. The authors focused on moving beyond short laboratory experiments by testing the robot in a more realistic in-dorm environment.

The study involved 11 college students with clinically significant ADHD symptoms. The researchers conducted a three-week within-subjects study where students used a computer-based system without the robot for one week and the robot study companion for two weeks. The robot called Blossom, was an inexpensive open-source 3D-printed robot connected with a webcam, Raspberry Pi 4, and touchscreen. The robot mainly used subtle idle behaviors such as gaze shifts, posture movements, and sigh-like movements rather than actively interrupting the students. The researchers found that 10 out of 11 participants voluntarily continued using the robot when its use was no longer required. The system also received an average System Usability scale score of about 83.8.



Figure 2: The Blossom robot, Logitech webcam, and 7-inch touch screen UI connected to a Raspberry Pi 4 computer



Figure 1: Blossom and the study system, including the tripod-mounted webcam and touch screen interface

Project Objective:

1. Evaluate whether college students with ADHD would find a SAR useful as a study companion.
2. Examine whether students would voluntarily continue using the robot when its use was no longer required.
3. Understand which robot features students liked or disliked during study sessions.
4. Investigate how students preferred the robot to behave during their study activities.
5. Identify additional features that could make the robot more useful.
6. Explore the feasibility of deploying a study companion robot for longer periods in students' real living environments.
7. Collect user feedback to guide the future design and development of the robot.

Research Gap:

Previous research had shown that socially assistive robots could support attention and provide non-verbal feedback, but many studies were conducted only during short laboratory sessions. The authors identified a need to understand how students would actually use such a robot over a longer period in their own living environment.

There were also relatively few studies investigating SARs specifically for college students with ADHD. Earlier systems such as Atten and Kip3 showed potential, but questions remained about long-term use and how robots could support students in more realistic situations. Another gap was the limited understanding of what students actually want from a study companion robot. The researchers therefore focused on collecting feedback about robot behavior, appearance, usability, and possible additional functions.

The paper also points toward future work involving visual and audio information to understand user behavior and predict the user's state during study sessions, which is particularly relevant to Focus Bot.

Research Questions:

- 1: Will college students with ADHD find a socially assistive robot useful as a study companion?
- 2: What features of the system do students like or dislike during study sessions?
- 3: How would students prefer the robot to behave during their study activities?
- 4: What additional features could make the robot study companion more useful to college students with ADHD?
- 5: Will students voluntarily continue using the robot when they are no longer required to use it?
- 6: How do the robot's physical movements and sounds affect the student's study experience?
- 7: How can future versions of the robot use visual and audio information to understand or predict the user's state during a study session?

Key Contribution:

The main contribution of the paper is the design and initial evaluation of an in-dorm socially assistive robot study companion for college students with ADHD. The researchers demonstrated that a relatively simple robot could still be perceived as useful by students. Participants reported that the robot created a feeling of companionship and accountability while its small size and appearance were also positively received. motor sounds and abrupt movements were reported as distracting by some participants

The study also identified several possible future features, including reminders, assignment tracking, improved portability, and assistant-like functions. The authors proposed future analysis of visual and audio data to better understand students' behavior and predict their state during study sessions.

Literature Review 04

Open-Source Robotic Study Companion with Multimodal Human–Robot Interaction to Improve the Learning Experience of University Students.

Overview:

This paper presents a Robotic Study Companion designed specifically for university students. The researchers focus on developing an affordable, customizable, and open-source desktop social robot that can provide students with an interactive learning experience.

The RSC was developed using cost-effective off-the-shelf components, with custom mechanical parts produced through 3D printing. The robot supports multimodal human robot interaction, including speech, gesture, touch, touchscreen interaction, light, and sound. The authors identify several problems faced by university students, including distractions, motivation problems, time-management difficulties, communication barriers, and mental-health-related challenges. They argue that existing educational robots are often expensive, difficult to customize, or designed mainly for younger students.

The RSC hardware cost was approximately and the authors released the design as an open-source project so that others can reproduce, modify, and improve the robot.



Project Objective:

1. Developing a social robot specifically suited to university students.
2. Making the robot affordable using readily available components.
3. Providing multiple interaction methods such as speech, touch, light, motion, and sound.
4. Supporting personalized and adaptable learning experiences.
5. Developing a compact and visually appealing desktop robot.
6. Making the hardware and software customizable and reproducible.
7. Providing an open-source platform that other researchers and students can modify and improve.

Research Gap:

The paper identifies that much of the existing research on social robots in education focuses on children and classroom environments, while the specific requirements of university students have received less attention. Another problem is that existing commercial social robots can be expensive, making them difficult for students and researchers to obtain. Some existing student-designed robots use affordable components, but they provide limited customization and lack features such as adaptive learning and broader academic assistance.

Research Questions:

- 1: How can an affordable social robot be designed specifically to support university students?
- 2: What interaction modalities should a robotic study companion provide to support effective human–robot interaction?
- 3: How can open-source hardware make a robotic study companion more affordable and customizable?
- 4: How can the robot provide personalized and adaptable learning support for different university students?

- 5: How can speech, gesture, touch, light, and sound be integrated into a compact desktop robot?
- 6: What hardware and software requirements are necessary for developing a reliable and user-friendly robotic study companion?
- 7: How can an open-source robotic platform allow other students and researchers to reproduce, customize, and improve the system?

Key Contribution:

The main contribution of this paper is the development of the Robotic Study Companion as an affordable, open-source desktop social robot. The researchers developed both the hardware and software, documented the design process, and released the project as an open-source platform. The robot uses 3D-printed components and off-the-shelf hardware to reduce cost and make future customization easier.

The RSC also provides several interaction methods. It can communicate through speech, gestures, touchscreen and RGB LED lighting, and sound. This multimodal approach allows different types of feedback and interaction during study sessions.

Literature Review 05

University Students' Acceptance of a Robot Study Companion

Overview:

This study investigates how university students perceive and accept the Robot Study Companion, an open-source tabletop social-educational robot. The researchers used the Technology Acceptance Model to examine students' perceived usefulness and perceived ease of use. Ten students from the University of Guyana participated in the study. Each participant first completed a pre-test questionnaire, interacted with the RSC prototype for approximately 10–15 minutes, and then completed a post-test questionnaire. The study examined students' opinions about the robot's usefulness, ease of interaction, satisfaction, and willingness to adopt it as a study tool. The participants also provided suggestions regarding voice clarity, interactivity, personalization, accessibility, and technical support.

Project Objective:

1. To evaluate how university students perceive the usefulness of the Robot Study Companion.
2. To evaluate the ease of use and interaction of the robot.
3. To identify factors that influence students' willingness to adopt a robot as a study tool.
4. To understand students' expectations regarding educational robots.
5. To identify possible improvements for future versions of the RSC.
6. To investigate the importance of personalization, accessibility, and AI integration.
7. To collect user feedback that can guide future development of the robot.

Research Gap:

The study identifies that although social robots are increasingly being explored in education, there is still limited understanding of how university students accept and interact with tabletop robotic study companions. Existing educational robots often focus on younger students, while the specific needs and expectations of university students remain less explored. The authors also identify adaptability and customization for university students as an open research area. In addition, there is a need to understand how features such as voice interaction, personalization, accessibility, and AI integration influence acceptance.

Research Questions:

1. How do university students perceive the usefulness of a robotic study companion?
2. How easy do students find the RSC to use and interact with?
3. What factors influence students' willingness to adopt the RSC as a study tool?

4. What features do university students expect from a robotic study companion?
5. How can personalization improve the usefulness of the RSC?
6. What improvements are needed in the robot's voice, interaction, and customization?
7. How can student feedback be used to improve future versions of the RSC?

Key Contribution:

The study provides user-based evidence about acceptance of a tabletop robotic study companion. The participants generally perceived the RSC as useful and easy to interact with. Overall satisfaction was reported on the studies scale and participants highlighted its interactive nature and potential to provide a focused learning experience. The study also identified voice clarity, personalization, accessibility, and technical support as areas for future improvement.

Literature Review 06

Co-Creating Buildable and Open Social Robot Study Companions with University Students

Overview:

This research focuses on improving the buildability, serviceability, and maintainability of open-source social robot study companions. The study develops the Robot Study Companion using a user-centered co-design process. The researchers applied the Double Diamond framework, together with Design for Assembly and Design for Disassembly principles. The

robot uses an affordable 3D-printed structure with components such as a Raspberry Pi, touchscreen, servo-driven flippers LED ring microphones, and speakers. The research involved students and developers from Guyana and Estonia.

Project Objective:

1. To improve the physical buildability of an open-source social robot.
2. To reduce the complexity of assembling the robot.
3. To make robot components easier to disassemble and repair.
4. To involve university students in the robot design process.
5. To apply Design for Assembly principles to the robot chassis.
6. To apply Design for Disassembly principles to improve maintenance.
7. To develop a more accessible and customizable study companion platform.

Research Gap:

The research points out that open-source robots can provide accessibility and customization, but the physical construction process itself can become a barrier. Existing robots may be supplied as pre-assembled systems, while open-source platforms can contain difficult fasteners, complicated wiring, and inaccessible service points. There was limited empirical evidence about whether mechanical redesign could make open social robots easier for first-time student builders to assemble and maintain.

Research Questions:

1. How can Design for Assembly principles improve the buildability of a social robot?
2. How can Design for Disassembly principles improve robot maintenance?
3. How do mechanical redesigns affect students' perceived workload?

4. How do redesigned components affect the usability of the robot?
5. Can the redesigned robot be assembled more easily by first-time builders?
6. How can open-source robot design support student participation and technical empowerment?
7. What improvements are still required in the robot's documentation and assembly guidance?

Key Contribution:

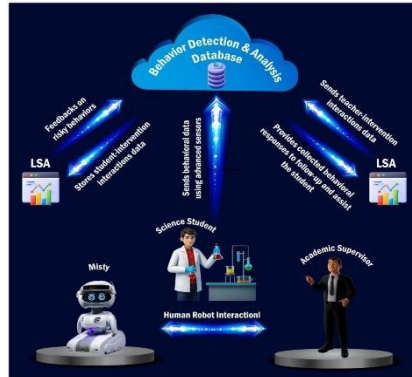
The researchers developed the RSC v4.1 with features such as twist-lock fasteners, snap-fit joints, and tool-free service access. Two studies were conducted: one with four RSC developers and another with seven first-time student builders. The reported SUS usability score for developers increased from 59.4 for v4 to 89.4 for v4.1 while mean developer assembly time decreased from 21.4 minutes to 13.7 minutes.

Literature Review 07

Socially Assistive Robot as Laboratory Safety Assistant for Science Students

Overview:

This study proposes a Socially Assistive Robot for laboratory safety assistance. The proposed system uses a Misty II robot to monitor science students during laboratory activities and identify potentially dangerous behaviors. Examples include not wearing safety glasses or running around the laboratory. The robot is intended to provide monitoring, assistance, companionship, and friendly supervision when a human laboratory supervisor is not immediately available. An analytics dashboard is also included as part of the intervention framework.



Project objects:

1. To develop a robot-assisted laboratory safety assistant framework.
2. To detect potentially dangerous student behaviors.
3. To use robot sensors and cameras to collect behavioral information.
4. To provide timely assistance when unsafe behavior is detected.
5. To provide friendly supervision when a human supervisor is unavailable.
6. To use an analytics dashboard as part of the intervention system.
7. To explore the use of socially assistive robots in educational environments.

Research Gap:

The paper notes that although social robots have been used in several intervention and educational applications, their use for laboratory safety monitoring and behavioral intervention remains limited. Laboratories contain different types of hazards, and students may sometimes perform unsafe activities. Traditional supervision depends heavily on human instructors or supervisors. Therefore, there is a need to explore how social robots equipped with sensing and interaction capabilities can support monitoring and intervention in educational laboratory environments.

Research Questions:

1. How can a social robot detect unsafe behaviors during laboratory activities?
2. What types of student behaviors should be monitored for laboratory safety?
3. How can robot sensors and cameras support real-time behavioral monitoring?
4. How can a social robot provide appropriate intervention when unsafe behavior is detected?
5. How can an analytics dashboard support robot-assisted safety monitoring?
6. How can social robots provide friendly supervision in the absence of a human supervisor?
7. How can robot-assisted monitoring be integrated into educational laboratory environments?

Key Contribution:

The major contribution is the proposal of a robot-assisted laboratory safety framework using Misty II. The framework combines sensing, behavioral monitoring, intervention, and an analytics

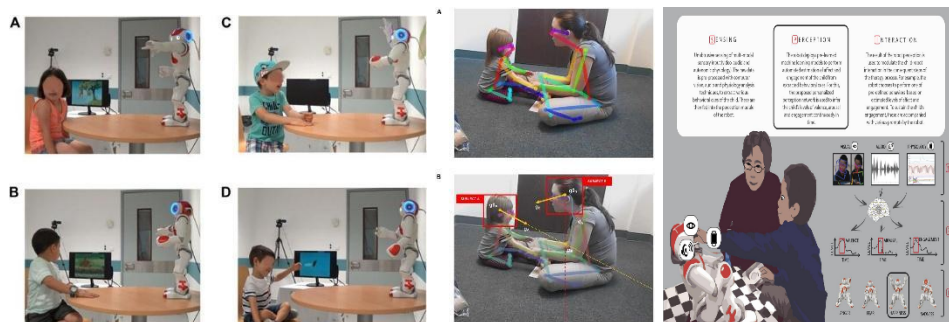
dashboard. The authors position the robot as a system that can monitor, assist, and provide companionship/friendly supervision to students.

IEEE Literature Review 08

Attention Analysis in Robotic-Assistive Therapy for Children with Autism

Overview:

This study focuses on automatically assessing the attention response of children with autism spectrum disorder during robotic-assisted therapy. The researchers aimed to develop a quantitative model that could evaluate attention during human–robot interaction. The motivation is that attention is important for learning and therapy, but manually assessing attention can be difficult and subjective. The paper therefore investigates computational methods for estimating attention using behavioral and interaction-related information.



Project Objective:

1. To develop a quantitative model for assessing attention during robotic-assisted therapy.
2. To automatically evaluate children's attention responses.
3. To investigate behavioral indicators related to attention.

4. To support therapists in monitoring attention during therapy sessions.
5. To provide quantitative information that can help evaluate behavioral progress.
6. To reduce reliance on purely manual attention assessment.
7. To explore the potential of robotic-assisted interaction for behavioral analysis.

Research Gap:

The paper states that previous attempts to quantify attention during human–robot interaction were limited. In particular, the researchers identify the need for a more quantitative and automated approach to evaluate attention responses during robotic-assisted therapeutic sessions. Manual observation can make continuous and objective assessment difficult. Therefore, the study explores computational attention analysis that can provide measurable information to therapists.

Research Questions:

1. How can a robot-assisted system automatically assess a user's attention?
2. What behavioral information can be used to estimate attention?
3. How can attention response be represented quantitatively?
4. How accurately can a computational model identify attention during human–robot interaction?
5. How can automated attention analysis support therapists?
6. How can continuous attention monitoring help evaluate behavioral progress?
7. What limitations should be considered when using automated attention analysis in real-world interaction?

Key Contribution:

The main contribution is the development of a quantitative attention-analysis approach for robotic-assisted therapy. Instead of relying only on subjective observation the study investigates

computational assessment of attention response during human robot interaction. This provides a foundation for using sensing and machine-learning-based analysis to understand user behavior.

2025 IEEE Literature Review 09

Motivating Students Self-study with Goal Reminder and Emotional Support

Overview:

This research investigates how a social robot can support university students during self-study. The robot acts as a peer study companion and provides two major types of support: goal reminders and positive emotional support. The researchers conducted an exploratory Wizard-of-Oz study and compared these behaviors with a control condition where the robot mainly provided physical presence. The study examined students' perceived focus, productivity engagement, ease of use, and willingness to use the robot again. The results suggest that goal reminders and emotional support can improve students experience with the robot with the goal-reminder condition showing higher willingness for future use .

Project Objective:

1. To investigate the role of a social robot as a self-study companion.
2. To examine whether goal reminders can support students during study sessions.
3. To investigate the effect of positive emotional support from a robot.
4. To evaluate students' perceived focus while interacting with the robot.
5. To evaluate students' perceived productivity and engagement.
6. To examine students' willingness to use the robot in future study sessions.

7. To explore how different robot behaviors affect the study experience.

Research Gap:

Although social robots have been investigated in educational settings, their role in supporting students during independent self-study has received less attention. In particular, there is limited research comparing different types of robotic support, such as goal reminders and emotional encouragement. The study therefore explores whether a robot can provide more meaningful support than simply being physically present during study sessions.

Research Questions:

1. How can a social robot support university student during self-study?
2. Do goal reminders help students maintain their study focus?
3. Does positive emotional support improve students' study experience?
4. How does robotic support affect students' perceived productivity?
5. How does robotic support affect students' engagement during self-study?
6. Which type of robotic support is more acceptable to students for future study sessions?
7. How does the robot's physical presence influence the student's self-study experience?

Key Contribution:

The study demonstrates the potential of a social robot to act as a peer study companion rather than simply providing physical presence. Goal reminders and emotional support were investigated as active forms of robotic assistance. The results showed positive effects on perceived ease of use, while the goal-reminder condition was associated with greater willingness to use the robot in future study sessions.

2025 34th IEEE Literature Review 10

Emotive Design of a Robot Study Companion to Support University Learning

Overview:

This research focuses on designing a robot study companion that can express emotions during interaction with university students. The researchers highlight that university students can experience low motivation, emotional fatigue, and academic stress. Although social robots have potential as study companions, many existing systems lack appropriate emotional expressivity and contextual interaction for university-level learning. The study therefore investigates how a robot can communicate emotions through multimodal behaviors such as facial expression movement, color, and vocal behavior.

Project Objective:

1. To design emotional expressions for a robotic study companion.
 2. To make the robot's interaction more natural and engaging.
 3. To investigate how robots can communicate emotions to university students.
 4. To use multiple modalities for emotional expression.
 5. To improve the interaction experience between students and the robot.
 6. To support motivation and engagement during university learning.
 7. To develop an emotionally expressive study companion suitable for higher education.
- ResearchGate

Research Gap:

Existing social robots used in education can provide information and interaction, but many lack rich emotional expression and contextual relevance for university students. The research identifies a need for study companions that can communicate emotions in a more natural and multimodal manner. This is particularly relevant because university students may experience motivation problems, emotional fatigue, and academic stress.

Research Questions:

1. How can a robot study companion express emotions during university learning?
2. Which emotional expressions are appropriate for university students?
3. How can facial expressions improve human robot interaction?
4. How can robot movement contribute to emotional communication?
5. How can lights or colors be used to communicate the robot's emotional state?
6. How can vocal behavior make the robots interaction more expressive?
7. How can multimodal emotional expression improve students engagement with a study companion?

Key Contribution:

The research contributes to the design of an emotionally expressive robot study companion for university learning. Instead of relying only on speech or basic interaction, the robot can use multiple communication channels, including facial expression movement color and vocal behavior. This provides a foundation for creating study companions that feel more interactive and socially engaging.

